Q1 JEE Main 2020 - 2 September (Morning)

In a reactor, $2\mathrm{kg}$ of ${}_{92}\mathrm{U}^{235}$ fuel is fully used up in 30 days. The energy released per fission is $200\mathrm{MeV}$. Given that the Avogadro number, $N=6.023\times 10^{26}$ per kilo mole and $1\mathrm{eV}=1.6\times 10^{-19}\mathrm{J}$. The power output of the reactor is close to

(A) $125\mathrm{MW}$
(B) 35 MW
(C) $60\mathrm{MW}$
(D) 54 MW

Q2 JEE Main 2020 - 3 September (Morning)

In a radioactive material, fraction of active material remaining after time t is $9/16$. The fraction that was remaining after $t/2$ is

(A) $\frac{3}{4}$
(B) $\frac{4}{5}$
(C) $\frac{3}{5}$
(D) $\frac{7}{8}$

Q3 JEE Main 2020 - 3 September (Evening)

The radius R of a nucleus of mass number A can be estimated by the formula $R=\left(1.3\times 10^{-15}\right)\mathrm{A}^{1/3}\mathrm{m}$. It follows that the mass density of a nucleus is of the order of $\left(\mathrm{M}_{\text{prot.}}\cong M_{\text{neut.}}=1.67\times 10^{-27}\mathrm{kg}\right)$

(A) $10^{24}\mathrm{kgm}^{-3}$
(B) $10^{10}\mathrm{kgm}^{-3}$
(C) $10^{17}\mathrm{kgm}^{-3}$
(D) $10^{3}\mathrm{kgm}^{-3}$

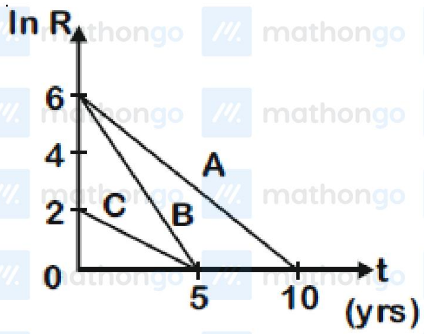
Q4 JEE Main 2020 - 4 September (Evening)

Find the binding energy per nucleon for ${}^{120}\mathrm{Sn}$.
Mass of proton $\mathrm{m_p}=1.00783\mathrm{U}$, mass of neutron $\mathrm{m_n}=1.00867\mathrm{U}$ and mass of tin nucleus $m_{sn}=119.902199$ U. (take $1\cup=931\mathrm{MeV}$)

(A) $9.0 \mathrm{MeV}$
(B) $8.5 \mathrm{MeV}$
(C) $8.0 \mathrm{MeV}$
(D) $7.5 \mathrm{MeV}$

Q5 JEE Main 2020 - 5 September (Morning)

Activities of three radioactive substances A, B and C are represented by the curves A, B and C, in the figure. Then their half-lives $\mathrm{T}_{\frac{1}{2}}(\mathrm{A}) : \mathrm{T}_{\frac{1}{2}}(\mathrm{B}) : \mathrm{T}_{\frac{1}{2}}(\mathrm{C})$ are in the ratio



(A) 02:01:03
(B) 04:03:01
(C) 02:01:01
(D) 03:02:01

Q6 JEE Main 2020 - 5 September (Evening)

A radioactive nucleus decays by two different processes. The half life for the first process is $10 \mathrm{s}$ and that for the second is $100 \mathrm{s}$. The effective half life of the nucleus is close to

(A) $12 \mathrm{sec}$
(B) $9 \mathrm{sec}$
(C) $55 \mathrm{sec}$
(D) 6 sec

Q7 JEE Main 2020 - 6 September (Morning)

You are given that Mass of ${}_3^7\mathrm{Li} = 7.0160\mathrm{u}$
Mass of ${}_2^4\mathrm{He} = 4.0026\mathrm{u}$
and Mass of ${}_1^1\mathrm{H} = 1.0079\mathrm{u}$
When $20\mathrm{g}$ of ${}_3^7\mathrm{Li}$ is converted into ${}_2^4\mathrm{He}$ by proton capture, the energy liberated, (in kWh), is
[Mass of nucleon $= 1\mathrm{GeV/c^2}$]

(A) 1.33×10^6
(B) 4.5×10^5
(C) 8×10^6
(D) 6.82×10^5

Q8 JEE Main 2020 - 6 September (Evening)

Given the masses of various atomic particles $m_p = 1.0072u, m_n = 1.0087u, m_e = 0.000548u$
$\mathrm{n} \equiv$ neutron, $\mathrm{e} \equiv$ electron, $\overline{\mathrm{v}} \equiv$ antineutrino and $\mathrm{d} \equiv$ deuteron. Which of the following process is allowed by momentum and energy conservation?

(A) $\mathrm{n} + \mathrm{n} \rightarrow$ deuterium atom
(electron bound to the nucleus)
(B) $\mathrm{n} + \mathrm{p} \rightarrow \mathrm{d} + \gamma$
(C) $\mathbf{p} \rightarrow \mathbf{n} + \mathbf{e}^+ + \bar{v}$
(D) $\mathrm{e}^+ + \mathrm{e}^- \rightarrow \gamma$

Q9 JEE Main 2020 - 7 January (Evening)

Activity of a substance changes from $700\ s^{-1}$ to $500\ s^{-1}$ in 30 minute. Find its half-life in minutes

(A) 66
(B) 62
(C) 56
(D) 50

Q10 JEE Main 2020 - 9 January (Evening)

Two gases-argon (atomic radius $0.07\ nm$, atomic weight 40) and xenon (atomic radius $0.1\ nm$, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to:

(A) 1.83
(B) 2.3
(C) 3.67
(D) 1.07

Answer Key

Q1 (C)	**Q2** (A)	**Q3** (C)	**Q4** (B)
Q5 (A)	**Q6** (B)	**Q7** (A)	**Q8** (B)
Q9 (B)	**Q10** (D)		

Q1

Power $= \frac{\text{Energy released}}{\text{time}}$

$= \frac{1}{30\times24\times60\times60}\left(\frac{2\times6.023\times10^{26}}{235}\ \times200\times1.6\times10^{-19}\right)\mathrm{MJ/s}$

$\simeq 60\mathrm{MW}$

Q2

$\mathrm{N}=\mathrm{N}_0\mathrm{e}^{-\lambda t}$

$\lambda t=\ln\frac{16}{9}$

$\mathrm{N}_1=\mathrm{N}_0\mathrm{e}^{-\lambda t2}$

$\mathrm{N}_1=\frac{3}{4}\mathrm{N}_0$

Q3

$V=\frac{4}{3}\pi r^3=\frac{4}{3}\pi r_0^3A$

$\rho=\frac{\mathrm{M}}{\mathrm{V}}=\frac{1.67\times10^{-27}\mathrm{A}}{\frac{4}{3}\pi\mathrm{r}_0^3\mathrm{A}}$

Q4

B.E. $=(\Delta\mathrm{m})\mathrm{C}^2$

$\Delta m=(50m_p+70m_n)-(m_{sn})$

$=(50\times1.00783+70\times1.00867)$

$-(119.902199)$

B.E. $=\Delta\mathrm{m}\times931\mathrm{MeV}$

$\Rightarrow$ B.E. $=1020.5631\mathrm{MeV}$

$\Rightarrow\frac{\text{B.E.}}{\text{Nucl.}}=\frac{1020.5631}{120}=8.5\mathrm{MeV}$

Q5

$R=\lambda N_0e^{-\lambda t}$

$\Rightarrow\ln R=\ln\lambda N_0-\lambda t$

$\Rightarrow$ Slope of curves $=-\lambda$

also $\lambda = \frac{\ln 2}{t_1/2}$

$$\left.\begin{array}{l} \text{for A slope } = -\frac{6}{10} \\ \text{for B slope } = -\frac{6}{5} \\ \text{for C slope } = -\frac{2}{5} \end{array}\right] \Rightarrow \mathrm{T}_{1/2}(\mathrm{A}) : \mathrm{T}_{1/2}(\mathrm{B}) : \mathrm{T}_{1/2}(\mathrm{C}) = 2:1:3$$

Q6

$\lambda_e = \lambda_1 + \lambda_2$

$\Rightarrow \frac{\ln 2}{\mathrm{T_e}} = \frac{\ln 2}{\mathrm{T_1}} + \frac{\ln 2}{\mathrm{T_2}}$

$\Rightarrow T_e = \frac{T_1 \times T_2}{T_1 + T_2} = \frac{10 \times 100}{110}$

$= 9\mathrm{sec}$

Q7

${}_3\mathrm{Li} + {}_1^1\mathrm{H} \longrightarrow 2{}_2^4\mathrm{He}$

$\Delta \mathrm{m} = (\mathrm{m_{Li}} + \mathrm{m_H} - 2\mathrm{m_{He}})$

$= .0187\mathrm{u}$

Q value $= \Delta \mathrm{mc}^2$

$= 17.42\mathrm{MeV}$

Energy liberated $= \frac{20}{7} \times 6.023 \times 10^{23} \times (\mathrm{Q}-$ value $)$

$= 300 \times 10^{29}\mathrm{eV}$

$= 480 \times 10^{10}\mathrm{J}$

$= 1.33 \times 10^{6}\mathrm{kWh}$

Q8

$\Delta \mathrm{m}$ should be positive

$(m_p + m_n) > m_d$

$\Rightarrow$ only (2) is possible

Q9

$$\ell n\left[\frac{A_0}{A_T}\right]=\lambda t$$

$$\Rightarrow \quad \ell n 2=\lambda t_{1/2} \qquad \ldots\text{(i)}$$

$$\Rightarrow \quad \ell n\left[\frac{700}{500}\right]=\lambda\,(30\,\text{min}) \quad \ldots\text{(ii)}$$

(i) / (ii)

$$\Rightarrow \quad \frac{\ell n 2}{\ell n\,(7/5)}=\frac{t_{1/2}}{(30\,\text{min})}$$

$$\Rightarrow \quad (2.06004)\,30=t_{1/2}=61.8\,\text{min.}$$

Q10

$$\text{Mean free time}=\frac{1}{\sqrt{2}n\pi d^2 v_{rms}}$$

$$\frac{t_{Ar}}{t_{Xe}}=\frac{d_{Xe}^2}{d_{Ar}^2}\times\sqrt{\frac{m_1}{m_2}}$$

$$=\left(\frac{0.1}{0.07}\right)^2\times\sqrt{\frac{40}{140}}$$

$$=\left(\frac{10}{7}\right)^2\times\sqrt{\frac{2}{7}}=1.07$$